Enhance Platform Engineering with an Internal Developer Platform

Learn how to navigate the CI/CD landscape with GitHub Actions, Argo, and Flux.

Software developers are always eager to leverage the latest technology to enhance their productivity and performance, and develop high-quality code at a speed never experienced before. Thanks to advancements in self-service internal platform engineering (IPE), it is now possible to bring continuous integration and continuous delivery (CI/CD) together. This practice automates the software development process, ensuring the seamless delivery of high-quality applications. Implemented correctly, it optimises the developer experience and accelerates the delivery of quality business applications. It also helps embrace artificial intelligence (AI), helping organisations compete while generating value.

Innovation and adoption of CI/CD tools have come full circle — from open source to commercial solutions and then to cloud hyperscale providers like AWS, GCP, and Azure building their own tools. Now, open source is at the forefront once again, especially with multi-cloud adoption presenting challenges in mastering proprietary platform tools. Let's look at how all of this can be simplified using a plug-and-play model, starting with user-friendly IDEs integrating with platform tools like GitHub Actions, Argo, and Flux.

All developers strive for stability at the core with a feature-rich, easy-to-use IDE that gives flexibility and agility at the edge, in order to connect to various critical endpoints for source control of orchestration and CI/CD pipelines. These pipelines automate the build of infrastructure across registry, compute, data, networking, service, observability, and security.

This may sound great, but the challenge is organising it into a scalable internal developer platform. This platform should seamlessly integrate with development environments to establish developer productivity quickly and reliably. The reference architecture diagram in Figure 1 breaks down plug-and-play dev tools for CI/CD across common tools. These can be adapted to your organisation's setup, providing the ability to work in the multi-cloud development and delivery landscape embraced by most organisations today.

Platform engineering is a hot trend, and is bringing together infrastructure development and software application development under the umbrella of DevOps. The standardisation of the internal developer platform has its benefits, as organisations constantly hire new developers who bring their own expertise on specific productivity tools, making them efficient and, more importantly, happy to contribute, collaborate, and stick with the firm longer. The plug-and-play model of the internal developer platform supports a developer's abilities to independently run, manage, and develop applications while complying with an organisation's best practices to ensure reliability and security.

How to get started

If you are just getting started on building an internal platform, begin by establishing reusable, composable, and configurable platform components. This will enhance productivity and help embrace standardisation. Consider treating the internal developer platform as a product where developers are the end users who identify and prioritise technical capabilities, tools, and processes that are most useful.